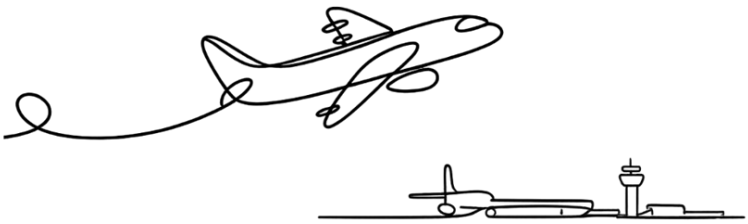
Airport Simulation User's Guide



Group 5

App Installation

Extract the contents of the zip file and run the main executable file

Loading Screen

As you open the app, you will be greeted with the loading screen. The loading screen consists of three buttons: 'Configure Simulation', 'View Simulation' and 'Compare Simulations'. This can also be seen in Figure 1.

Figure 1: Loading Screen

Configure Simulation

By selecting this option, you enter the airport configuration menu. This screen allows you to model an airport of your choice, as long as the inputs are valid.

Parameters include:

- **'Generational Model'** - This allows the user to choose the model they want to use for aircraft generation (between Uniform and Poisson).
- **'Inbound Flow Rate'** - This refers to the number of aircraft per hour that will enter the holding pattern of the airport (capped at 1000).
- **'Outbound Flow Rate'** - This refers to the number of aircraft per hour that will enter the take-off queue of the airport (capped at 1000).
- **'Emergency probability'** - This allows you to configure the probability of an emergency occurring on an inbound plane. (Default is 0%)

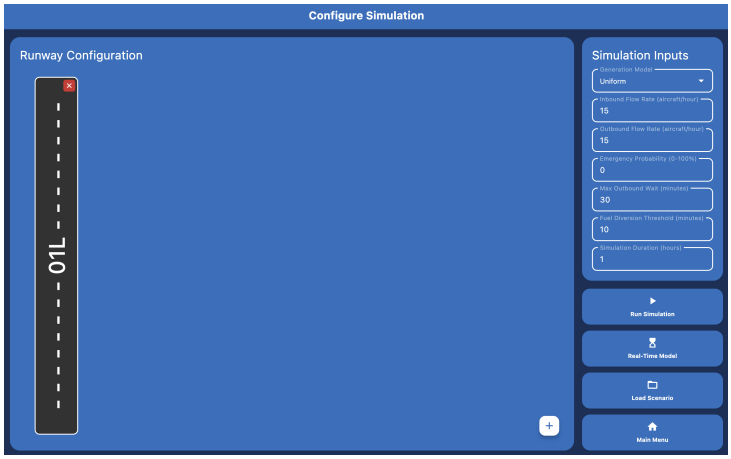


Figure 2: Configure Simulation Screen

- **'Max Outbound Wait'** - This is the maximum time that a departing plane waits in the take-off queue before it is cancelled (minimum 1 minute). (Default is 30 mins)
- **'Fuel Diversion Threshold'** - The minimum amount of fuel that a plane can hold before it has to declare an emergency landing (minimum 1 minute). (Default is 30 mins)
- **'Simulation Duration'** - This allows you to configure how many hours you want the simulation to be for. (Default and minimum is 1 hour).

Runways Configuration

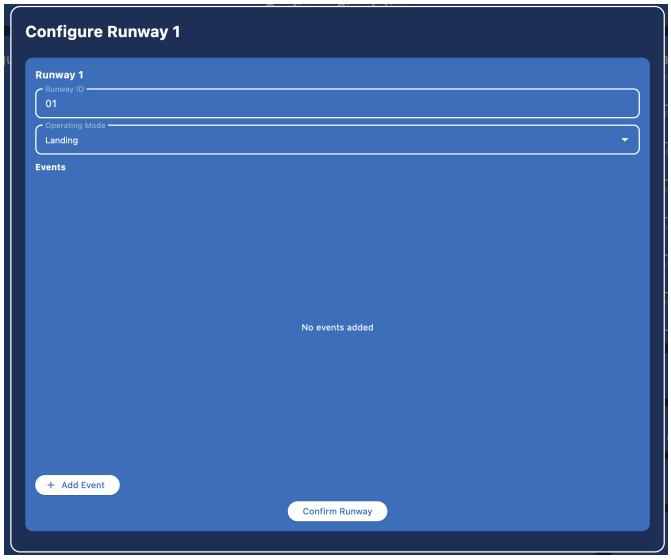


Figure 3: Runway Configuration

This section allows you to configure the runways in the airport.

Adding Runways:

The '+' box allows the user to add up to 10 runways to the airport.

Configuring Individual Runways:

In order to configure each runway, the user must right-click on each runway and set its operating mode (defaulted to 'Landing Mode') and the 'Runway ID'.

Adding Runway Events:

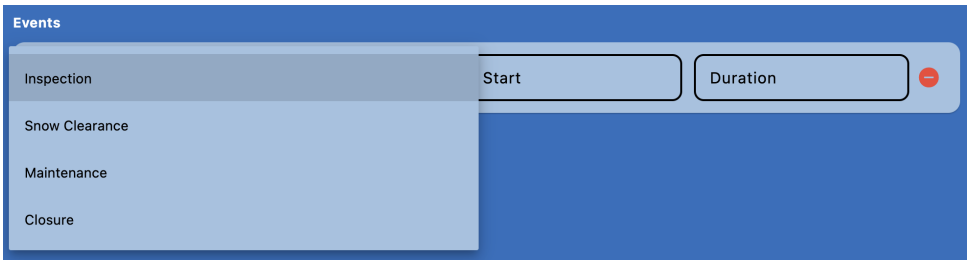


Figure 4: Runway Events Configuration

In the runway configuration screen, the user has the ability to add runway events. In this, they must configure each event by selecting the 'Event Type', 'Start Time' and 'Duration'. Once a runway event has been added, it is possible to delete it by pressing the red button to the right of the event box.

Important Notes: It is important for the user to fill in all required input fields. If this is not done, the simulation will not start. The user will be prompted to fill in the missing fields which will be highlighted and change colour.

Once all the required inputs are successfully filled, the user has two options to choose from on how they would like to run the simulation: 'Run Simulation' and 'Real-Time Model'.

Run Simulation

Using 'Run Simulation' will mean that all the processing will be done in the background and the user will be provided all the required statistics from the simulation along with graphical representations of some of the data.

On the results screen, the user is given multiple options for what they want to do with this simulation configuration: 'Export to CSV', 'Save Scenario', 'Load Scenario' and 'Configuration Screen'. Check out the 'View Results' section for more information.

Real-Time Model

The 'Real-Time Model' allows the user to have control over the airport's runways in real-time. This means that they can schedule events and runway closures through the

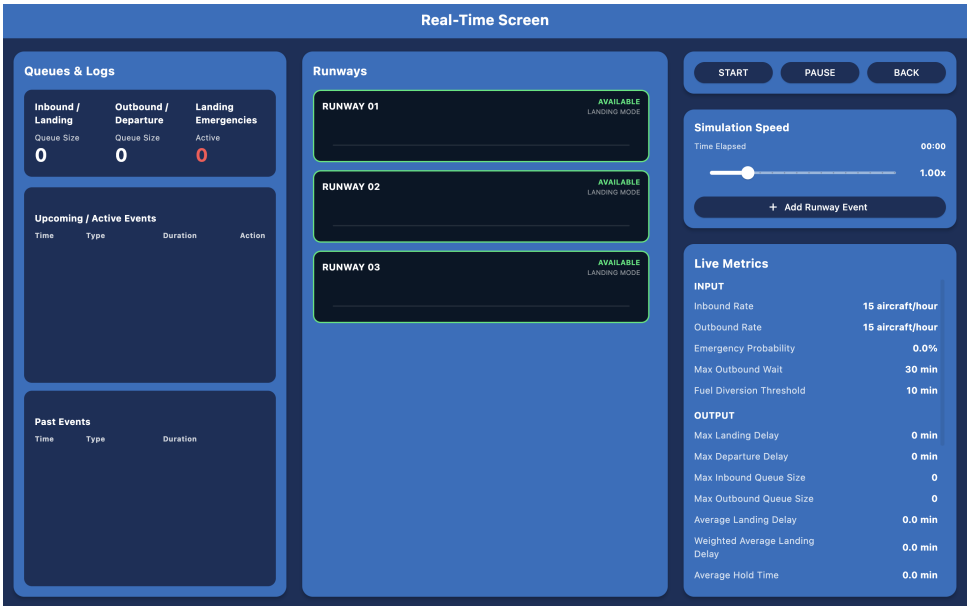


Figure 5: Real-Time Model Interface

duration of the simulation. This has the same functionality as 'Run Simulation' but provides the user with a more interactive interface.

When adding a runway event in the real-time run, the user is prompted to enter the type of runway event, its start time and the duration of the event, similar to the core simulation.

Cancelling Runway Events:

In order to cancel a runway event, the user must press the red button on the right side of the event in the runway event log.

Speeding Up the Simulation:

The simulation speed can be controlled using the slider on the top right of the screen beneath the start, pause and back buttons.

Simulation Controls:

- **Pause:** This button pauses the simulation. To restart the simulation, press the start button.
- **Start:** This starts the simulation and the simulation timer.
- **Back:** This button will take you back to the airport configuration screen.

Once the 'Real-Time' Simulation ends, the user enters the results screen which provides the user with the statistics from the simulation along with their graphical representations.

View Saved Scenario

The 'View Saved Scenario' button allows the user to view and compare previously saved simulations. This screen allows users to compare simulations which they have saved. The user can load in saved simulations using the 'Load Scenario 1' and 'Load Scenario 2' buttons on the top-left of the screen, above the table. The user will then be prompted to load a scenario into a table. Once both scenarios have been loaded, the comparison graphs will automatically appear on the right side of the screen.

View Results

The 'View Results' button allows the user to view the detailed results of an individual simulation run that has been previously saved.

The user is given multiple options for what they want to do with this simulation configuration:

- **Export to CSV:** This will generate a CSV file of the selected simulation.
- **Save Scenario:** This button will save the simulation to the system's database.
- **Load Scenario:** This button will load and show the results screen for a simulation that was stored on the database.
- **Configure Simulation:** This button will allow the user to re-configure the input values of the scenario, allowing for the generation of a new simulation.